

Intelligent traffic light control

The Traffic Light Controllers (TLC) used now are based on microcontrollers and microprocessors. This TLC uses predefined hardware, and so can only function according to the program and does not have the intelligence to be flexible on a real-time basis. Ever had to wait at a crossroad when there is absolutely no vehicles incoming from the adjoining roads but still you have to wait for the timer to count-down to zero because someone thought that was a pretty neat idea? Ever found yourself philosophising about your life because there is nothing really else to do except inhale exhaust fumes? All this in addition to the fact that there is so much petrol being wasted for absolutely nothing. That horrid experience could soon never have to be experienced by anyone again. Now there has been a proposed system of Intelligent Traffic Light controller which can be used to give the system flexibility of modification on a real-time basis. This new method will use sensor networks coupled with embedded technology to act smarter. Here, first the total traffic on all the connected roads on a crossing will be noted. After considering this, the timings of the red and green signals (no one really cares about orange signals, it seems) will be appropriately adjusted so as to ensure maximum efficiency in the movement of the vehicles to prevent major traffic jams.

Automotive technologies

Embedded systems are contributing to a safer system of driving. Every automobile manufacturer which comes out now uses embedded technology to make their cars sweeter. So what can we expect from cars eventually? Embedded systems can regulate speeds, just in case you're prone to putting your foot down and trying to be a speed demon, and the car will try and maintain safe distance to the car in front of you, and in some cases even intelligently make decisions based on traffic conditions, weather (and hopefully how drunk you are). Imagine that, the end of drunk driving and road rage? Currently, embedded systems in automobiles are rugged, as they are usually made up of just a single chip. Because of this ruggedness, there is a much reduced chance of failure, and almost no possibility that the system will get busy with another process and hang. This also makes the whole system small enough to fit inside a car's nooks and crannies.

One of the more exciting applications of embedded technology is that soon you could see the first driverless car (insert that driverless car show joke here), although it wouldn't be that cool, or that lame depending on how